

ESD (Emergent Structure Dynamics)

Trade Structure Analysis using Scale and Effective Diversity

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ESD (Emergent Structure Dynamics) is a framework for transforming trade data into a dynamic state-space representation.

Instead of focusing only on export magnitude, ESD describes an economic system through:

- Scale
- Structural diversity
- Temporal movement

The framework converts trade tables into state vectors and observes how systems evolve through time.

Motivation

Traditional trade analysis typically focuses on:

- total export value
- growth rate
- market share
- trade balance

These metrics explain magnitude but often ignore structural evolution.

For example:

Two countries may have equal export volume:

Country A:

- exports evenly across many countries
- exports many products

Country B:

- exports mainly one product
- exports mainly one market

Total export values may be identical, but their structures differ significantly.

ESD aims to capture this hidden structural information.

Data

Source:

- UN Comtrade

Example dataset:

- Korea export
- 2010–2024
- HS2 level

Input variables:

Variable	Description
refYear	Year
partnerDesc	Export partner
cmdCode	HS code
primaryValue	Export value

State Definition

ESD transforms trade data into a dynamic state vector representation.

For a given year t :

$$\mathbf{S}_t = (L1_t, EffN_{country,t}, EffN_{HS,t})$$

where:

Variable	Meaning
$L1_t$	Total export scale
$EffN_{country,t}$	Effective number of export destinations
$EffN_{HS,t}$	Effective number of export products

Interpretation:

- $L1$ captures economic scale
 - $EffN_{country}$ captures market diversity
 - $EffN_{HS}$ captures product diversity
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Scale Definition

Scale is defined as:

$$L1_t = \sum_i x_i$$

where:

- x_i = export value

Interpretation:

- larger value → larger economic activity

Diversity Definition

Structural diversity is measured using HHI:

$$HHI = \frac{\sum_i x_i^2}{(\sum_i x_i)^2}$$

Effective Number:

$$EffN = \frac{1}{HHI}$$

Interpretation:

- High EffN
evenly distributed structure
- Low EffN
concentrated structure

Dynamic State Space

A sequence of yearly state vectors creates a trajectory:

$$\mathbf{S}_1 \rightarrow \mathbf{S}_2 \rightarrow \dots \rightarrow \mathbf{S}_n$$

Distance between states:

$$D_t = ||\hat{\mathbf{S}}_t - \hat{\mathbf{S}}_{t-1}||$$

where:

$$\hat{\mathbf{S}}$$

is a normalized state vector.

Interpretation:

- Direction indicates structural change
- Distance indicates magnitude of change
- Small movement → stable structure
- Large movement → structural transition or external shock

Examples:

Higher:

$$EffN_{country}$$

→ broader export markets

Lower:

$$EffN_{HS}$$

→ increasing product concentration

Korea Case Study (2010–2024)

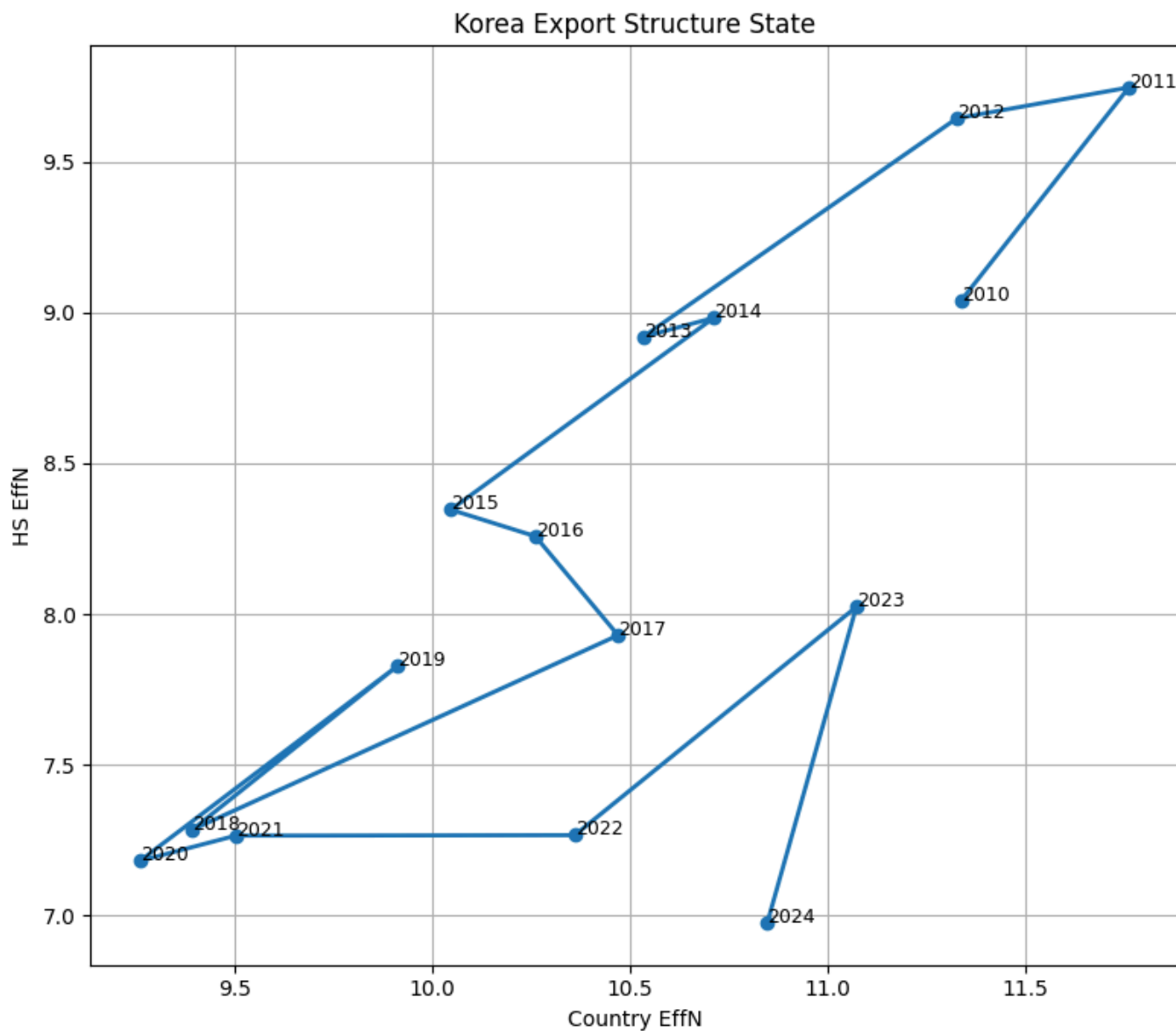
The framework was applied to Korea export data.

Analysis procedure:

1. Download UN Comtrade data
2. Aggregate yearly export tables
3. Calculate:
 - total scale ($L1$)
 - country diversity ($EffN_{country}$)
 - product diversity ($EffN_{HS}$)
4. Construct state trajectories
5. Analyze structural evolution

Results

Export state trajectory



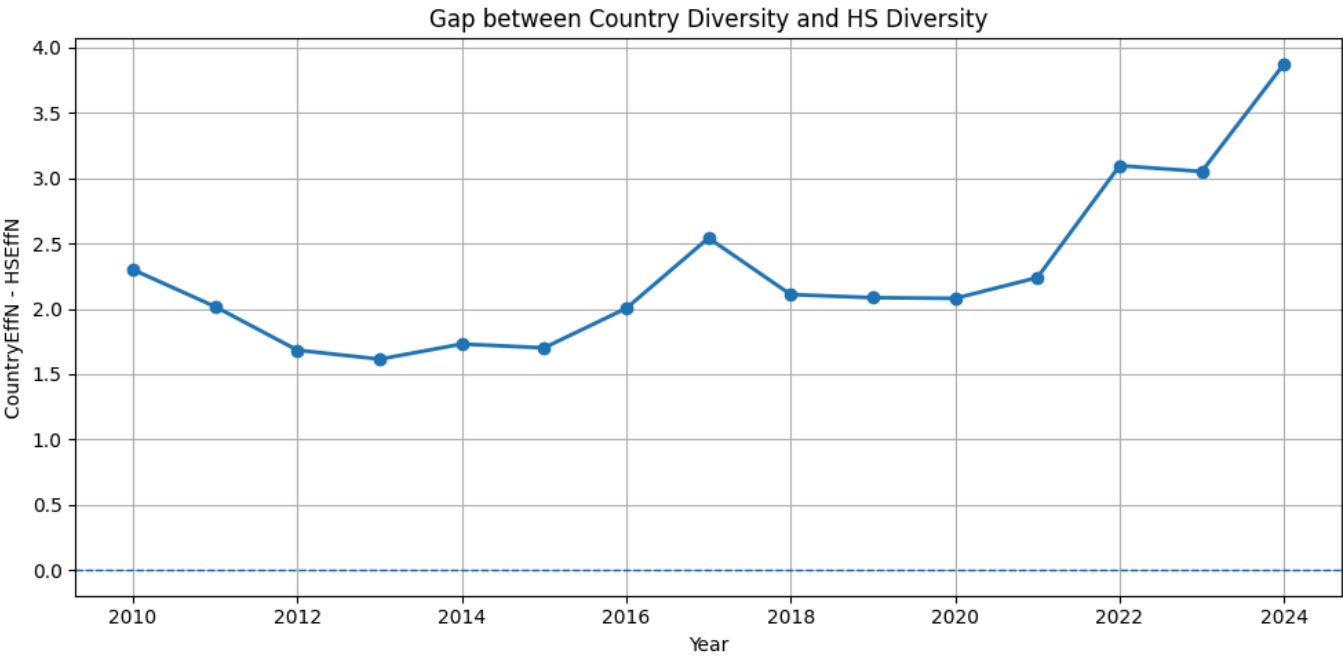
Observation:

- Early years show relatively high product diversity
- Later years show decreasing product diversity
- Country diversity remains relatively stable

Diversity gap

Defined as:

$$Gap = EffN_{country} - EffN_{HS}$$



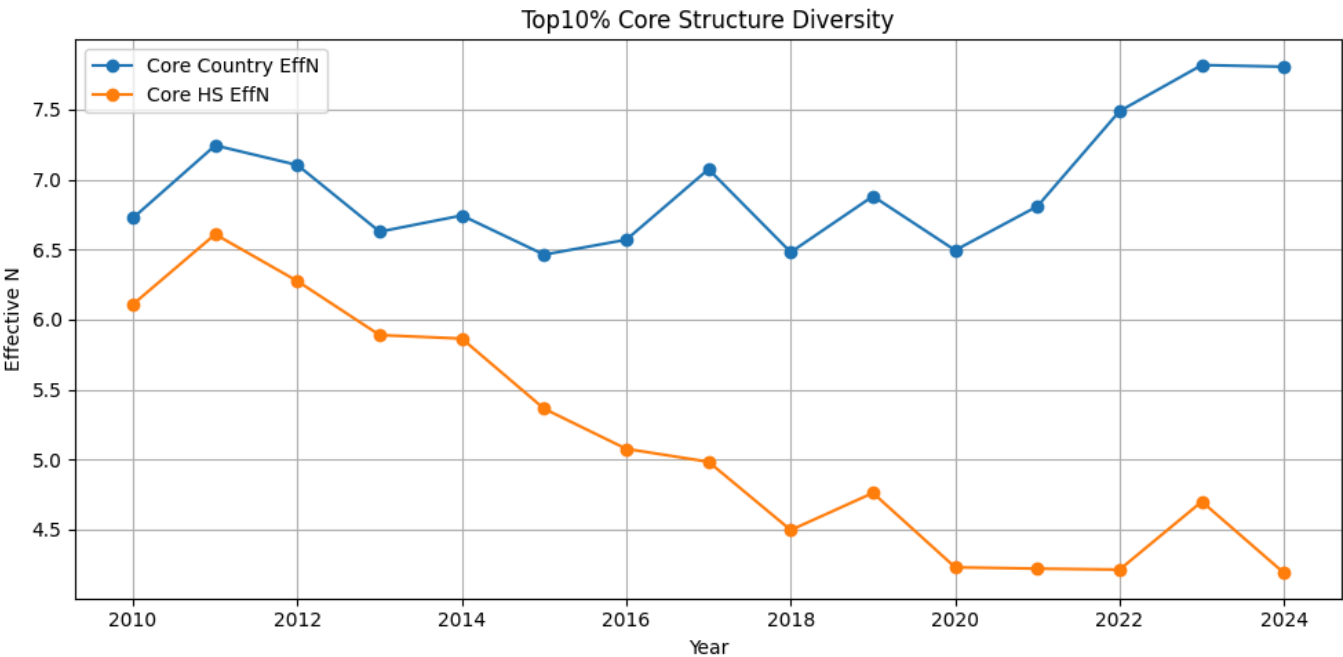
Observation:

Gap increases over time.

Interpretation:

Korea exports to many countries, but through fewer dominant products.

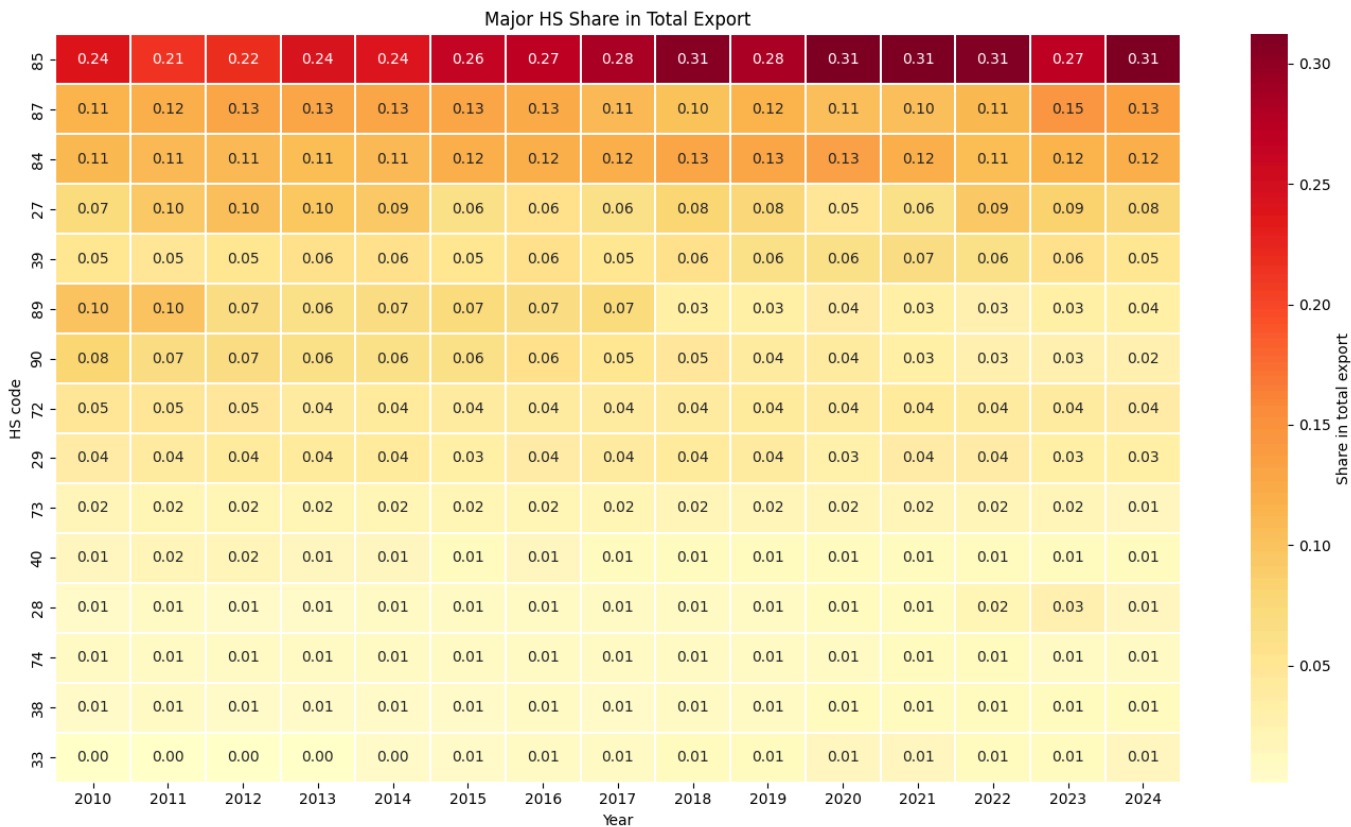
Core structure (Top 10%)



Observation:

Core product diversity decreases continuously.

Major product concentration



Observation:

HS85 becomes increasingly dominant.

HS85 partner structure



Observation:

Production networks gradually shift toward broader Asian supply chains.

Framework Insights

- 1. Scale and diversity are independent dimensions.
- 2. Country diversity and product diversity evolve differently.
- 3. Structural concentration can increase even when total trade grows.
- 4. State trajectories reveal hidden structural transitions.

Future Work

Potential extensions:

- Wasserstein trajectory distance
- Density fields (KDE)
- Energy interpretation
- State transition probabilities
- ESD latent dynamics

Data Sources and References

UN Comtrade

United Nations Comtrade Database

- Main database: <https://comtrade.un.org/>
- API documentation: <https://comtradedeveloper.un.org/>

Country Mapping

R package: `comtradr`

Reference:

https://docs.ropensci.org/comtradr/reference/country_codes.html

Data Access

Trade data can be programmatically queried through:

- Python API
- R `comtradr` package
- bulk download services